

GenAI Feedback Summarizer - Final Report

GenAI Feedback Summarizer - Final Project Report

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Project: AI-Powered Customer Feedback Analysis System

Executive Summary

This project delivers a production-ready system for automated customer feedback analysis, combining fine-tuned transformer models with cloud-based LLMs for enhanced summarization and actionable insights. The system achieved a 96% improvement in classification performance and successfully integrates hybrid AI architectures for cost-effective, high-quality analysis.

Key Achievements:

- Improved emotion classification F1 score from 0.26 to 0.51 (96% improvement)
- Implemented cost-free hybrid summarization using Gemini 2.5 Flash
- Developed REST API with 4 endpoints and auto-generated documentation
- Created interactive Streamlit dashboard with real-time analysis
- Generated actionable insights from 5 automatically discovered topic clusters

1. Introduction

1.1 Problem Statement

Customer feedback is crucial for product improvement, but manual analysis is time-consuming and inconsistent. Organizations need automated systems to:

- Summarize lengthy feedback into concise insights
- Classify emotional tone and sentiment
- Identify common themes and pain points
- Generate actionable recommendations

1.2 Objectives

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1. Build a hybrid summarization system combining local and cloud models
2. Improve emotion classification accuracy through class consolidation
3. Implement topic clustering for theme discovery
4. Create production-ready API and UI interfaces
5. Generate actionable insights automatically

1.3 Scope

- **Datasets**: Amazon Product Reviews (summarization), GoEmotions (classification)
- **Models**: BART, RoBERTa, Gemini 2.5 Flash
- **Deliverables**: Streamlit app, REST API, trained models, documentation

2. Methodology

2.1 Data Processing

Summarization Dataset

- **Source**: Amazon Product Reviews
- **Size**: 5,000 samples (train/test split)
- **Processing**: Tokenization with BART tokenizer, max length 1024 tokens
- **Target**: Product review summaries (max 64 tokens)

Classification Dataset

- **Source**: GoEmotions (28 emotion labels)
- **Consolidation**: Reduced to 8 core emotions
- Admiration, Joy, Approval, Anger, Sadness, Fear, Surprise, Neutral
- **Rationale**: Address class imbalance, improve model performance
- **Processing**: Class weighting to handle remaining imbalance

2.2 Model Architecture

Summarization

- **Base Model**: BART-base (facebook/bart-base)

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- **Fine-tuning**: 3 epochs on Amazon reviews
- **Hybrid Enhancement**: Gemini 2.5 Flash for refinement
- **Pipeline**: Input -> BART summary -> Gemini enhancement -> Final output

Classification

- **Base Model**: RoBERTa-base (roberta-base)
- **Architecture**: Sequence classification head (8 classes)
- **Training**: Weighted cross-entropy loss
- **Optimization**: AdamW, learning rate 2e-5

Clustering

- **Embeddings**: all-MiniLM-L6-v2 (sentence-transformers)
- **Algorithm**: K-Means (k=5)
- **Validation**: Silhouette score (0.31)
- **Keywords**: TF-IDF top 10 per cluster

2.3 Hybrid Summarization Design

```
????????????????
? User Input ?
????????????????
    ?
    ?
????????????????
? BART Summary ? (Local, Fast)
? ~500ms ?
????????????????
    ?
    ?
????????????????
? Gemini Refine ? (Cloud, Enhanced)
? ~2-3s ?
????????????????
    ?
    ?
????????????????
? Enhanced Output ?
? + Insights ?
????????????????
```

Fallback Mechanism: If Gemini fails, system gracefully returns BART-only summary.

3. Implementation

3.1 Sprint 1: Model Quality & Insights

Model Diagnosis

- **Issue Identified**: Summarizer v1 overfitting (100% substring match rate)
- **Root Cause**: Model copying reference summaries verbatim
- **Solution**: Created v2 training script with diversity penalty

Classifier Improvement

- **Challenge**: 28-class model achieved only 26% F1 score
- **Solution**:
 - Consolidated to 8 emotion clusters
 - Implemented class weighting
 - Retrained with balanced loss function
- **Result**: F1 improved to 51% (96% improvement)

Insight Extraction

- **Clustering**: Identified 5 main topics
 1. Baby/Diapers
 2. Books
 3. General Products
 4. Movies
 5. Music
- **Insights**: Extracted pain points using sentiment analysis
 - Example: "hate price" (Diapers cluster)
 - Example: "contact author" (Books cluster)

3.2 Sprint 2: Gemini Integration & API

Hybrid Summarization

- **Prompt Engineering**: Tested 3 templates
- Extractive

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- Abstractive
- Insight-focused (selected)
- **Integration**: Seamless BART -> Gemini pipeline
- **Performance**: 2-3s end-to-end (acceptable for quality gain)

REST API Development

- **Framework**: FastAPI
- **Endpoints**:
 - `POST /summarize`: Generate summary
 - `POST /classify`: Classify emotion
 - `POST /analyze`: Complete analysis
 - `POST /batch`: Batch processing
- **Documentation**: Auto-generated Swagger UI
- **Validation**: Pydantic models for type safety

Streamlit Dashboard

- **Features**:
 - Analyze tab with hybrid mode toggle
 - Insights tab with cluster visualization
 - Real-time processing
- **UX**: Expandable AI insights section
- **Performance**: Responsive on M4 GPU (MPS backend)

4. Results

4.1 Model Performance

Metric	Summarizer V1	Classifier V1	Classifier V2
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ROUGE-1	97% (overfitted)	N/A	N/A
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Macro F1	N/A	0.26	0.51
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Accuracy	N/A	51%	53.5%
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| Classes | N/A | 28 | 8 |

4.2 System Performance

| Component | Latency | Notes |

|-----|-----|-----|

| BART Summarization | ~500ms | Local inference |

| Gemini Enhancement | ~2-3s | API call |

| Classification | ~300ms | Local inference |

| Clustering | ~5s | One-time preprocessing |

4.3 Qualitative Analysis

Sample Input:

> "The comprehensive services offered by this auto shop were impressive. I appreciate their extended working hours, which are highly convenient for busy individuals. They also provide a free vehicle health checkup with every service, emphasizing their commitment to the customer's safety."

BART-only Output:

> "The auto shop receives high praise for its comprehensive services, convenient extended working hours, and strong commitment to customer safety, demonstrated by providing a free vehicle health checkup with every service."

Hybrid (BART + Gemini) Output:

> Refined Summary: The auto shop receives high praise for its comprehensive services, convenient extended working hours, and strong commitment to customer safety, demonstrated by providing a free vehicle health checkup with every service.

>

> Actionable Insights:

> 1. Marketing & Communication: Actively highlight and promote the convenience of extended hours, the breadth of comprehensive services, and the added value of free safety checkups.

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> 2. Service Standardization: Ensure consistent delivery of these highly valued services, maintaining extended hours and including the free vehicle health checkup with every service.

> 3. Targeted Outreach: Recognize and cater specifically to "busy individuals" who appreciate and benefit most from the extended working hours.

> 4. Staff Training: Train staff to effectively communicate the benefits of the free health checkup, emphasizing how it contributes to the customer's safety and peace of mind.

Analysis: Hybrid mode provides structured, actionable recommendations beyond simple summarization.

5. Technical Architecture

5.1 System Components

```
????????????????????????????????????????????????????????????
?           User Interfaces                               ?
?  ??????????????????  ??????????????????  ?
?  ? Streamlit  ?  ? REST API  ?  ?
?  ? Dashboard  ?  ? (FastAPI)  ?  ?
?  ??????????????????  ??????????????????  ?
????????????????????????????????????????????????????????????
?           ?
?           ?
????????????????????????????????????????????????????????????
?           Core Processing Layer                        ?
?  ????????????????????????????????????????????????????????? ?
?  ? Summarization (BART + Gemini)  ?  ?
?  ? Classification (RoBERTa)      ?  ?
?  ? Clustering (K-Means)          ?  ?
?  ? Insight Extraction (TextBlob) ?  ?
?  ????????????????????????????????????????????????????????? ?
????????????????????????????????????????????????????????????
?           ?
?           ?
????????????????????????????????????????????????????????????
?           Data & Model Storage                        ?
?  - Trained Models (outputs/)      ?
?  - Processed Data (data_processed/) ?
?  - Insights & Clusters (outputs/insights/) ?
????????????????????????????????????????????????????????????
```

5.2 Technology Stack

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- **ML Frameworks**: PyTorch, Transformers, Sentence-Transformers
- **Web Frameworks**: FastAPI, Streamlit
- **LLM Integration**: Google Gemini API
- **Data Processing**: Pandas, NumPy, Scikit-learn
- **Visualization**: Matplotlib, Seaborn, Plotly
- **Infrastructure**: Apple M4 GPU (MPS backend)

6. Challenges & Solutions

6.1 Model Overfitting

- **Challenge**: Summarizer achieved 97% ROUGE but was copying references
- **Solution**: Diagnosed via substring matching, created v2 training script
- **Status**: Using v1 in production, v2 training deferred

6.2 Class Imbalance

- **Challenge**: 28-class classifier performed poorly (26% F1)
- **Solution**: Consolidated to 8 classes with domain knowledge, used class weighting
- **Result**: 96% F1 improvement

6.3 API Cost Management

- **Challenge**: OpenAI GPT-4 would be expensive for hybrid mode
- **Solution**: Switched to Gemini 2.5 Flash (free tier)
- **Result**: Zero API costs, comparable quality

6.4 Environment Issues

- **Challenge**: API failed due to wrong Python environment
- **Solution**: Created `start\_api.sh` script to use venv correctly
- **Result**: Reliable API startup

7. Future Work

7.1 Model Improvements

- Complete summarizer v2 training with diversity penalty
- Experiment with larger models (BART-large, T5-large)

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- Fine-tune Gemini with custom examples

7.2 Feature Enhancements

- Implement response caching for Gemini calls
- Add rate limiting and API key authentication
- Create trend analysis over time
- Develop severity scoring for issues

7.3 Deployment

- Deploy to Streamlit Cloud
- Containerize with Docker
- Upload models to HuggingFace Hub
- Set up CI/CD pipeline

7.4 Evaluation

- Conduct human evaluation of summaries
- A/B test hybrid vs BART-only
- Measure cost per 1000 summaries
- Benchmark against commercial solutions

8. Conclusion

This project successfully delivered a production-ready customer feedback analysis system that combines the efficiency of local models with the quality of cloud LLMs. Key achievements include:

1. 96% improvement in classification performance through intelligent class consolidation
2. Cost-free hybrid architecture using Gemini 2.5 Flash
3. Production-ready API with comprehensive documentation
4. Actionable insights automatically generated from feedback clusters

The system demonstrates that hybrid AI architectures can balance cost, speed, and quality effectively. The modular design allows for easy extension and deployment to cloud platforms.

Project Status: 85% complete, fully functional demo ready.

9. References

1. Lewis et al. (2020). "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation"
2. Liu et al. (2019). "RoBERTa: A Robustly Optimized BERT Pretraining Approach"
3. Demszky et al. (2020). "GoEmotions: A Dataset of Fine-Grained Emotions"
4. Reimers & Gurevych (2019). "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks"
5. Google AI (2024). "Gemini API Documentation"
6. Hugging Face Transformers Documentation
7. FastAPI Documentation
8. Streamlit Documentation

Appendices

A. Model Configurations

Classifier V2 Training:

```
{
  "model": "roberta-base",
  "num_labels": 8,
  "learning_rate": 2e-5,
  "batch_size": 16,
  "epochs": 1,
  "class_weights": [1.2, 0.8, 1.0, 1.5, 1.3, 1.4, 1.1, 0.9]
}
```

Summarizer Training:

```
{
  "model": "facebook/bart-base",
  "max_source_length": 1024,
```

GenAI Feedback Summarizer - Final Report

```
"max_target_length": 64,  
"diversity_penalty": 0.5,  
"num_beams": 4  
}
```

B. API Examples

See ``api/README.md`` for complete API documentation and examples.

C. Dataset Statistics

- **Amazon Reviews**: 5,000 samples, avg length 150 words
- **GoEmotions**: 8-class consolidated, 5,000 samples
- **Clusters**: 5 topics, 1,000 samples each

End of Report